2024 International Conference on Intelligent and Innovative Technologies in Computing, Electrical and Electronics (IITCEE) | 979-8-3503-0641-5/24/$31.00 ©2024 IEEE | DOI: 10.1109/IITCEE59897.2024.10467762

Clinical Text Classification Using Sequential Forward Selection

Rajkumar Rajasekaran
Department of Analytics
Vellore Institute of Technology
Vellore, India
vitrajkumar@gmail.com

Veer Sanghavi
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
veersanghavi8@gmail.com

Rashi Solanki
School of Computer Science and Engineering
Vellore Institute of Technology
Vellore, India
rashisolanki2002@gmail.com

Jolly Masih
Department of Business Analytics
BML Munjal University
Gurugram, India
jolly.iabm@gmail.com

***Abstract*—Clinical text classification is a critical undertaking within the realm of Natural Language Processing, holding significant implications for healthcare applications. In this NLP project, our primary objective is the categorization of medical transcripts into their corresponding medical conditions. To accomplish this, we harness the power of Sequential Forward Selection (SFS), a feature selection technique meticulously chosen for its capacity to streamline data dimensions. By employing SFS, our project aspires to not only enhance classification performance but also elevate the efficiency of pattern recognition, ensuring both speed and accuracy in disease identification. This research contribution underscores the pivotal role of Clinical Text Classification, with a specific focus on optimizing the process using SFS.**

Keywords— Clinical Text Classification, Medical transcripts, Sequential Forward Selection

I. Introduction

Clinical text classification stands as a critical undertaking within the healthcare domain, driven by the need to harness the wealth of information contained in unstructured clinical text data. This field has gained prominence due to its multifaceted importance and potential impact on healthcare management and patient outcomes.

In this NLP project, we propose a method that can accurately classify medical transcripts into their respective medical specialties. The target variable is the medical specialty and the features are the text data extracted from the medical transcripts. The project will involve several key steps. The text data will need to be pre-processed, which may involve tasks such as tokenization, stemming, and stop word removal. Once the data has been pre-processed, various machine learning algorithms are trained and evaluated on the dataset, such as Logistic regression, Support Vector Machines, or Categorical Boosting. The performance of each model can be evaluated using metrics such as accuracy, precision, recall, and F1-score. Finally, the model with the best performance can be selected and deployed to classify new medical transcripts into their respective medical specialties.

II. Data Description

We acquired the Medical Transcriptions dataset from Kaggle, which was obtained through the scraping of content from mtsamples.com.

TABLE I. DETAILED DATASET DESCRIPTION

Column Names	Missing Values	Missing Value %	Unique Values	Column Definition
Description	0	0	2348	Short description of transcription
medical specialty	0	0	40	Medical specialty classification of transcription
sample name	0	0	2377	Transcription title
Transcription	33	1	2358	Sample medical transcriptions
Keywords	1068	21	3848	Relevant keywords from transcription

III. Methodology

In the realm of Natural Language Processing (NLP) projects, a structured approach to data preparation and model development is paramount. This section outlines the key steps involved in the preparation of data for analysis, providing a clear framework for researchers and practitioners alike.

Data cleaning constitutes the initial phase and involves a comprehensive assessment of the dataset. This process aims to rectify issues such as missing values, duplicates, or inconsistent formatting. The integrity of the data is crucial for subsequent analysis, and diligent cleaning ensures its reliability.

Preprocessing follows data cleaning and focuses on transforming raw text data into a suitable format for analysis. Tasks encompassed in this stage encompass tokenization, POS (Part-of-Speech) tagging, and lemmatization. These steps facilitate the conversion of textual information into a structured and analyzable form.

Model data preparation entails the division of the dataset into distinct subsets, including training, validation, and testing sets. Furthermore, it involves the conversion of textual data into a numerical format compatible with machine learning models. This step is crucial for model training and evaluation.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 15,2024 at 05:46:18 UTC from IEEE Xplore. Restrictions apply.

Feature selection is a critical process involving the identification of the most relevant attributes from the dataset. These attributes serve as the basis for machine learning models. Techniques for feature selection may range from selecting the most frequent words to employing statistical methods that identify predictive features. This step enhances model efficiency by focusing on pertinent data.

The final step, model building, revolves around the training and evaluation of machine learning models. This phase encompasses various model types, such as classification or regression models, depending on the research objectives.

In summary, the meticulous execution of these methodological steps is fundamental to the success of NLP projects. They guarantee the accuracy and reliability of the data utilized, as well as the efficacy of the resulting machine learning models in addressing the research questions or objectives.

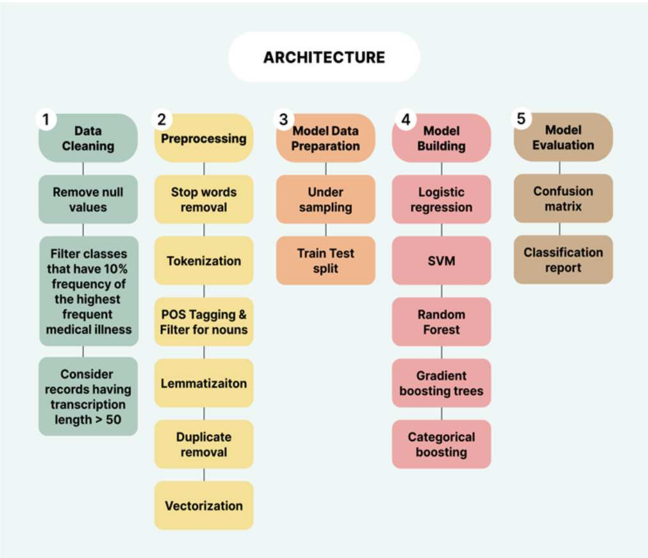

Fig. 1. Detailed Workflow of the Model

IV. High Level Solution Flow

A. Data Cleaning

When working on an NLP project, data cleaning is an important step to ensure that the data used for analysis is accurate and reliable. One common issue in datasets is the presence of null values, which can affect the performance of NLP models. To address this, the null values need to be removed from the dataset. This has be done by identifying the rows or columns that contain null values and either dropping these rows. Once the null values have been removed, the dataset is further processed by filtering classes that have records that 10% of the maximum number of records of the respective class in the dataset. Later the records are filtered by choosing only those records that have the transcription length more than 50.

B. Preprocessing

1) Stop words removal from nltk library and domain specific as well: Stop words are commonly occurring words that are often removed from text during NLP preprocessing as they are considered to have little impact on the overall meaning of a sentence. The Natural Language Toolkit (nltk) library provides a list of commonly occurring stop words that can be removed from text data. Additionally, domain-specific (in out case medical stop words) stop words can also be created and removed from the text data to improve the performance of NLP models.

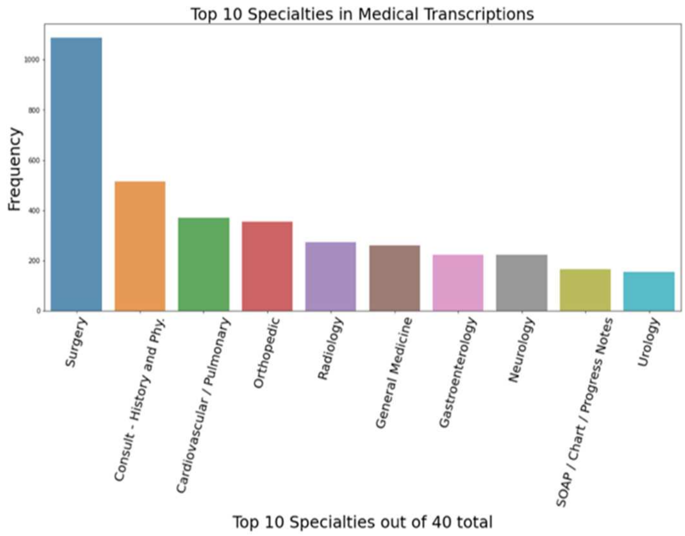

Fig. 2. Ten most Frequent Medical Illnesses in the Dataset

2) Tokenization using nltk library: Tokenization is the process of breaking down a sentence or text into smaller units, called tokens. The nltk library provides various methods for tokenization, such as word tokenize, which splits text into individual words, and sent tokenize, which splits text into sentences.

3) POS tagging and filtering for nouns using nltk library: It is the process of identifying and labeling the parts of speech in a sentence. The nltk library provides tools for performing POS tagging, which can be used to filter for specific parts of speech, such as nouns. This can be useful for identifying the most relevant words in a text for further analysis.

4) Lemmatization using WordNetLemmatizer: Lemmatization is the process of reducing a word to its base or root form, which can be useful for reducing the dimensionality of text data and improving the performance of NLP models. The WordNetLemmatizer provided by the nltk library can be used to perform lemmatization on text data.

5) Duplicate removal from list of features: Duplicate features can lead to overfitting and decreased model performance. Therefore, it is important to remove duplicates from the list of features. This can be done using various methods, such as using set() function to convert the list into a set and then back into a list, or by using pandas drop duplicates() function.

6) Vectorization using TfidfVectorizer: Vectorization is the process of converting text data into numerical form that is used in ML models. The TfidfVectorizer can be used to perform vectorization on text data, using the term frequency-inverse document frequency (TF-IDF) method, which gives more weight to less common words in the text data. This can improve the performance of NLP models.

Authorized licensed use limited to: Zhejiang University. Downloaded on October 15,2024 at 05:46:18 UTC from IEEE Xplore. Restrictions apply.

Fig. 3. Word Cloud for Medical Transcripts

C. Model Data Preparation

1) Train test split: We divide the data into two groups: one for practice and one for the actual test. This helps us see how well the model would perform on new data it hasn't seen before, making sure it isn't just memorizing the practice set but actually learning generalizable patterns.

2) Under sampling using RandomUnderSampler: Under sampling is a technique used in machine learning to address class imbalance, which occurs when one class in the dataset is much more prevalent than the others. This can lead to biased models that perform poorly on the minority class. RandomUnderSampler is a method provided by the imbalanced-learn library in Python that can be used to randomly remove samples from the majority class until the class distribution is more balanced. This can improve the performance of the model on the minority class.

D. Feature Selection using Forward Feature Selection

It is the process of identifying and selecting the most important features from a dataset that will be used in a machine learning model. This can improve model performance by reducing the dimensionality of the dataset, removing noise and irrelevant features, and improving the model's interpretability.

Forward feature selection is a method of feature selection that involves starting with an empty set of features and iteratively adding one feature at a time based on the improvement in model performance. The process starts by training a model on each individual feature and selecting the one that results in the highest performance. Then, the process is repeated by training models with all possible combinations of the previously selected features and selecting the combination that results in the highest performance. This process is repeated until a desired level of performance or a predefined number of features is reached.

Forward feature selection has several advantages, such as being computationally efficient and providing a simple way to identify the most important features in a dataset. However, it can also suffer from overfitting if the number of selected features is too large compared to the size of the dataset. Therefore, it is important to perform cross-validation to ensure that the selected features generalize well to unseen data.

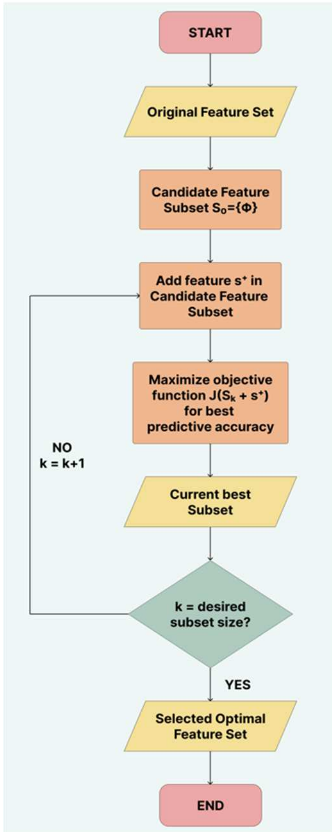

Fig. 4. Flowchart for Sequential Forward Selection (Feature Extraction Algorithm)

E. Model Building

1) Logistic regression: It is a statistical model where the goal is to predict whether a certain outcome will occur or not. It works by modeling the relationship between a set of independent variables. Logistic regression can also be used for multi-class classification problems.

2) SVM: It is a type of supervised learning algorithm used for classification and regression analysis. SVMs work by finding a hyperplane in a high-dimensional space that best separates the data points into different classes. SVMs are known for their ability to handle non-linearly separable data by using non-linear kernel func- tions, which transform the input data into higher-dimensional feature spaces.

3) Random Forest: It is a popular ensemble learning method used for classification, regression and feature selection tasks. It works by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes or mean prediction of the individual trees. Random Forest can handle a large number of input features and is robust to noisy data and outliers.

4) Gradient Boosting trees: It is a type of boosting algorithm that combines multiple weak models into a strong model by iteratively training each model to correct the errors of the previous model. Gradient Boosting trees is an extension of Gradient Boosting that uses decision trees as the weak models. The algorithm works by iteratively adding decision trees to the model, where each tree is trained on the residual errors of the previous trees. Gradient Boosting trees can be used for classification and regression tasks and is known for its ability to handle complex and non-linear

Authorized licensed use limited to: Zhejiang University. Downloaded on October 15,2024 at 05:46:18 UTC from IEEE Xplore. Restrictions apply.

relationships between the input features and the target variable.

5) Categorical Boosting: Categorical Boosting is a type of Gradient Boosting algorithm that is specifically designed for categorical data. Categorical Boosting works by using decision trees with categorical splits instead of continuous splits, and by incorporating categorical encoding techniques such as one-hot encoding and target encoding. Categorical Boosting can be used for classification tasks and is known for its ability to handle high-cardinality categorical features and imbalanced datasets.

V. Experimentation

- For the Data processing we experimented with a few techniques to reduce the amount of data so that it can be easy for feature selection, where we came up with a method to only extract nouns from the transcripts with the help of POS Tagging.
- We experimented with different number of classes and recorded how the results were to arrive at certain conclu- sions.
- For the feature selection we experimented with different number of features, like 10 features was underfitting whereas 20 features was overfitting and that's why we came to a point where we considered 15 features.
- Later we tried to different models like Logistic Regres- sion, SVM, Random Forest, Gradient Boosting trees and Categorical Boosting to see which models works the best for this problem statement.
- We performed Cosine and Jaccard similarity where we observed that most of the transcriptions are the same mapped to different specialties.

VI. Results

This section should present the results of the project, including any insights gained and how they relate to the original business problem.

- We concluded the project by saying that the data is very less and would require more number of records to make precise predictions.
- We have achieved 99% accuracy for the two medical illnesses: 'Surgery' and 'Consultation and History' by using Categorical Boosting.

	precision	recall	f1-score	support
Surgery	0.98	1.00	0.99	149
Consult - History and Phy.	1.00	0.98	0.99	161
accuracy			0.99	310
macro avg	0.99	0.99	0.99	310
weighted avg	0.99	0.99	0.99	310

Fig. 5. Classification Report for 2 Medical Illnesses using Categorical Boosting

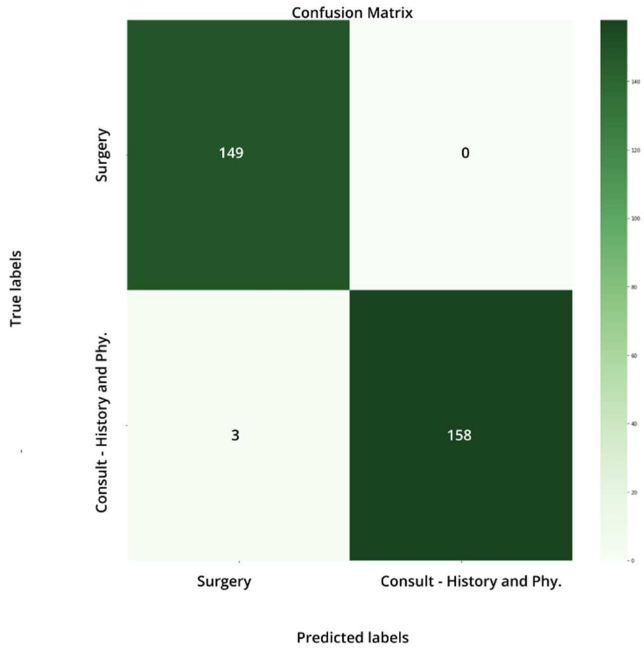

Fig. 6. Confusion Matrix (Heatmap) for 2 Medical Illnesses using Categorical Boosting

- Whereas, the accuracy is 75% when we choose three medical illnesses: 'Surgery', 'Consultation and History' and 'Cardiovascular/Pulmonary' by using Categorical Boosting. The reason for misclassification is that the records are being mapped to classes of Surgery and Consultation and History because for many of the other classes, their transcripts are describing a surgery about the medical specialty or describing the patient history. For example, cardiovascular and pulmonary being a target class has transcripts where a heart surgery is being described or the patient's heart history has been mentioned.

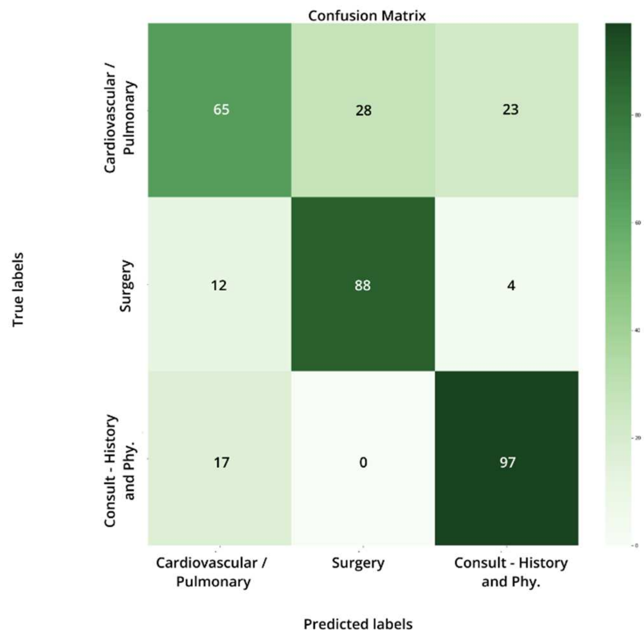

Fig. 7. Confusion Matrix (Heatmap) for 3 Medical Illnesses using Categorical Boosting

Authorized licensed use limited to: Zhejiang University. Downloaded on October 15,2024 at 05:46:18 UTC from IEEE Xplore. Restrictions apply.

	precision	recall	f1-score	support
Cardiovascular / Pulmonary	0.69	0.56	0.62	116
Surgery	0.76	0.85	0.80	104
Consult - History and Phy.	0.78	0.85	0.82	114
accuracy			0.75	334
macro avg	0.74	0.75	0.74	334
weighted avg	0.74	0.75	0.74	334

Fig. 8. Classification Report for 3 Medical Illnesses using Categorical Boosting

VII. CONCLUSION

We can use our knowledge of the subject matter to group similar categories together, which will reduce the total number of categories we need to consider. Creating custom features by hand might make the dataset perform better, but these custom features may not work as well for other transcription datasets that are not like this one. We have arrived at a conclusion that we would require more data to accurately classify the transcriptions to their respective medical categories. The future work may contain splitting the string in such a way that multiclass classification can be performed.

Our analysis shows that we need more data to accurately classify transcriptions into medical categories. The current dataset is too small, and this has limited our ability to achieve the desired accuracy. To address this, in the future, we will divide the transcriptions into smaller, more specific units. This will allow us to use a multiclass classification approach, which will enable us to categorize the medical content in a more nuanced and granular way. We believe that this will significantly improve the accuracy of our classification results, making our medical transcription system more useful and reliable. In simpler terms, we need to collect more data and divide it into smaller, more specific pieces. This will help us to more accurately classify the transcriptions into the correct medical categories.

REFERENCES

[1] Yao, L., Mao, C. & Luo, Y. Clinical text classification with rule-based features and knowledge-guided convolutional neural networks. BMC Med Inform Decis Mak 19 (Suppl 3), 71 (2019).

[2] Ravi, D., Wong, C., Deligianni, F., Berthelot, M., Andreu-Perez, J., Lo, B., & Yang, G. Z. Deep Learning for Health Informatics. IEEE journal of biomedical and health informatics, 21(1), 4–21, (2017).

[3] A. Marcano-Cedeño, J. Quintanilla-Domínguez, M. G. Cortina-Januchs and D. Andina. Feature selection using Sequential Forward Selection and classification applying Artificial Metaplasticity Neural Network. IECON 2010 - 36th Annual Conference on IEEE Industrial Electronics Society, Glendale, AZ, USA, pp. 2845-2850, (2010).

[4] Garla, V., Taylor, C., & Brandt, C. Semi-supervised clinical text classification with Laplacian SVMs: an application to cancer case management. Journal of biomedical informatics, 46(5), 869–875, (2013).

[5] Demner-Fushman, D., Chapman, W. W., & McDonald, C. J. What can natural language processing do for clinical decision support?. Journal of biomedical informatics, 42(5), 760–772, (2009).

[6] Özlem Uzuner, Ira Goldstein, Yuan Luo, Isaac Kohane. Identifying Patient Smoking Status from Medical Discharge Records. Journal of the American Medical Informatics Association, Volume 15, Issue 1, Pages 14–24, January 2008.

[7] Lance De Vine, Guido Zuccon, Bevan Koopman, Laurianne Sitbon, and Peter Bruza. Medical Semantic Similarity with a Neural Language Model. In Proceedings of the 23rd ACM International Conference on Conference on Information and Knowledge Management (CIKM '14). Association for Computing Machinery, New York, NY, USA, 1819–1822, (2014).

[8] Last, M., Kandel, A., & Maimon, O. Information-theoretic algorithm for feature selection. Pattern Recognition Letters, 22(6-7), 799-811, (2001).

[9] Kudo, Mineichi & Sklansky, Jack. Sklansky, J.: Comparison of Algorithms that Select Features for Pattern Classifiers. Pattern Recognition 33, 25-41. Pattern Recognition. 33. 25-41. (2000).

[10] D. Xiao and J. Zhang. Importance Degree of Features and Feature Selection. 2009 Sixth International Conference on Fuzzy Systems and Knowledge Discovery, Tianjin, China, pp. 197-201. (2009)

Authorized licensed use limited to: Zhejiang University. Downloaded on October 15,2024 at 05:46:18 UTC from IEEE Xplore. Restrictions apply.